

REWEAR-FUSED Engine A — Polymer-Chemistry Validation Dossier

TL;DR

- **Engine A is chemically plausible as an in-silico shortlisting exercise but cannot — and should not — claim a validated drop-in spandex replacement.** The locked design constraints (aliphatic diisocyanate + 25–35 wt% hard segment + polyester/polycarbonate soft segment ~2000 g/mol with embedded cleavable ester/carbamate in the amorphous phase) are each individually supported by primary polymer literature and are the scientifically correct levers to pull. The combination is genuinely novel and not yet demonstrated in a spun fiber.
 - **The single hardest scientific tension is "high stretch AND fast enzymatic cleavability at the same time."** Aliphatic-PU TPUs reaching 500–800% elongation exist in the literature (Špírková 2009; Kim 2022 for MDI vs H12MDI), and polyester-PU enzymatic degradation reaching 40.5% film weight loss in 12 h exists (CpCut1, AEM 2024) — but neither has been shown in the **same** dry-spun elastane filament. Engine A should claim only "predicted Tg + predicted elastomeric class + designed cleavage site," not validated cyclic-fatigue or recovery.
 - **ML-screening with PolyID + polyBERT + an XGBoost PU-Tg model is defensible for Tg ($R^2 \geq 0.91$, RMSE ≈ 9.6 K) and rough elastomeric classification, NOT for cyclic fatigue / elastic recovery / morphology / spinnability.** The honest framing — constraint-guided enumeration $\neq$ generative design $\neq$ wet-lab validation — is the right framing and is consistent with what each model's authors say in print.
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Key Findings

1. **Segmented PU-urea is the correct chemical platform.** Commercial spandex is a polyether-polyurea (PTMEG + MDI + diamine chain extender) whose elasticity arises from microphase separation between a low-Tg amorphous soft phase and hydrogen-bonded, often semicrystalline urea hard domains (Castagna et al. 2012, *Macromolecules* 45:8438; Versteegen et al. 2005, *Macromolecules* 38:3176; CAMEO/MFA fiber database).
2. **Aliphatic-only PU elastomers can hit >400% elongation but tend to have lower tensile strength than aromatic MDI analogues.** All-aliphatic HDI poly(carbonate-urethane)s give EB 600–800% (Špírková et al. 2009, *Polymer*); H12MDI-based bio-TPUs give 489–977% EB vs 11.07–30.69 MPa TS for the MDI analogue (Kim et al. 2022, *J. Polym. Res.* 29:467, DOI 10.1007/s10965-022-03375-x). (Springer)

3. **Aromatic isocyanates (MDI/TDI) are the right thing to exclude for an infant garment.** They degrade under physiological/thermal conditions to MDA and 2,4-TDA, which are IARC/NTP-listed potential carcinogens; the EUROPUR CertiPUR™ program limits aromatic diamines in PU foams to <5 mg/kg (Karlsson, Spence, Plehiers 2023, *Toxicol. Ind. Health* 39:298, DOI 10.1177/07482337231172816). (Sage Journals)
4. **Polyester (PCL) and polycarbonate soft segments are the right enzymatically-cleavable choice.** PCL has $T_g \approx -60^\circ\text{C}$, is hydrolyzable by lipases/cutinases/esterases, and PCL-based TPEs show 39.7 MPa UTS and 47.6% biodegradability versus 23.4 MPa and 24.3% for PTMEG TPE analogues (Park et al. 2023, *Polymers* 15:3209, DOI 10.3390/polym15153209). Polycarbonate diols (Placel CD220, 2000 g/mol) yield TPUs with hydrolytic stability superior to polyadipates and elastic recovery comparable to PTMEG-PCL at 300% strain (Gantrade/Daicel technical data). (nih) (Gantrade)
5. **Hard-segment content is the dominant lever for the stretch/strength trade-off.** Increasing HS content monotonically raises modulus and TS while reducing EB and hydrolysis resistance (Pugar et al. 2022; Ding et al. 2023). The closest matching peer-reviewed quantitative example is Bian et al. 2024 (*Int. J. Polym. Sci.* DOI 10.1155/2024/9248135), reporting TDI/MDI ratio changes give a 15.8% TS increase at a 7/3 ratio — confirming the qualitative direction. For spandex-like elongation, the literature consistently places HS content in the 25–40 wt% window; Engine A's 25–35 wt% target is in the validated regime. (Wiley Online Library)
6. **The cleavable bond MUST be in the amorphous phase.** Crystallinity is the dominant kinetic obstacle to enzymatic hydrolysis: PETase activity drops ~10-fold from amorphous (2–3% Xc) film to high-crystallinity ($\approx 45\%$) PET bottle resin (Cui et al. 2022, *Nature Commun.* 13:7180, DOI 10.1038/s41467-022-34908-z).
7. **Real, peer-reviewed enzymatic depolymerization of polyester-PU exists.** CpCut1 (*Cladosporium* sp. P7) degrades 40.5% of TPU film and 20.6% of post-consumer PU foam in 12 h at 55°C , with complete depolymerization of Impranil DLN-SD (Liu et al. 2024, *Appl. Environ. Microbiol.* 90:e01477-23, DOI 10.1128/aem.01477-23). UMG-SP1/SP2/SP3 hydrolyze >90% of TDA-DEG dicarbamate in 24 h (Branson et al. 2023, *Angew. Chem.* 62:e202216220, DOI 10.1002/anie.202216220). Engineered UMG-SP2 A141G shows >30.7-fold improved activity on polyester-PUR versus WT (Li et al. 2025, *Adv. Sci.* 12:2416019, DOI 10.1002/advs.202416019). (ResearchGate + 5)
8. **The HONEST limits of the ML screening must be stated up front.** PolyID's published mean absolute error for T_g is 19.8°C (test) and 26.4°C (in-house experimental set) (Aubuchon et al. 2023, *Macromolecules*, DOI 10.1021/acs.macromol.3c00994). polyBERT is trained on a hypothetical-polymer dataset built by BRICS-decomposing exactly 13,766 known polymers into 4,424 fragments (Kuenneth & Ramprasad 2023, *Nat. Commun.* 14:4099, Fig. 1c) — explicit, validated property prediction for segmented block PU-ureas is outside its tested domain. XGBoost on PU T_g (ACS Appl. Polym. Mater. 2025, DOI 10.1021/acsapm.5c04524) reaches $R^2 > 0.91$, RMSE $\approx 9.6\text{ K}$; on EB/TS, Ding et al. 2023 (*Chin. J. Polym. Sci.* DOI 10.1007/s10118-022-2838-6) reach R^2 0.85–0.91 on a curated/filtered

benchmark only after removing 49.6% of the raw dataset (R^2 0.73–0.78 on the full raw set).

None of these models predicts cyclic fatigue, true elastic recovery, hysteresis, or spinnability. (nih + 4)

9. **The realistic combinatorial design space is screenable.** 3 aliphatic diisocyanates $\times$ 4–6 cleavable polyols $\times$ 5 chain extenders $\times$ 5 hard-segment wt% bins $\approx$ 1,500–4,500 candidates — readily enumerable in RDKit and far smaller than the 100 million hypothetical polymers polyBERT was deployed on. RDKit enumeration of step-growth PU-urea repeat units is scientifically routine.
 10. **The strongest residual risks are spinnability and cyclic fatigue.** No peer-reviewed primary publication of a true dry-spun HDI-only, H12MDI-only, or IPDI-only spandex filament with simultaneously reported elongation + TS + elastic recovery was located in the searched literature. Engine A must not promise spinnability; it can only flag candidates predicted to be in the right elastomeric class.
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Details

PART 1 — Monomer Combinatorics & Design Space (chemistry validated)

1a. Segmented PU-urea elastomer architecture

A segmented thermoplastic polyurethane-urea elastomer consists of alternating soft segments (oligomeric polyol, MW 1000–3000 g/mol) and hard segments (diisocyanate + short-chain diol or diamine chain extender). The hard-soft thermodynamic incompatibility plus strong bidentate hydrogen bonding between adjacent urea N-H and C=O groups drives microphase separation into nanometer-scale hard domains (often semicrystalline) embedded in a continuous low-T_g amorphous soft phase. The hard domains act as thermo-reversible physical crosslinks; the soft phase carries the elongation. This nanoscale-segregated morphology is "well known to be key to determining their performance" (Castagna, Pangon, Choi, Dillon, Runt, *Macromolecules* 45:8438–8444, 2012, DOI 10.1021/ma3016568). Polyureas microphase separate even more strongly than polyurethanes because urea has higher cohesive energy (bidentate H-bonds — PMC9054001). (ScienceDirect + 2)

For spandex specifically: ≥ 85 wt% segmented polyurethane (FTC definition), typically PTMEG ≈ 1800 –2000 g/mol + MDI + diamine chain extender (EDA or hydrazine), with hard-segment content ~ 25 –40 wt%. Elongation 400–800% with elastic recovery $\approx 99\%$ at 200% strain is the commercial benchmark. (Google Patents + 3)

1b. Aliphatic diisocyanates — HDI, H12MDI, IPDI

Diisocyanate	Structure	Reactivity vs MDI	Light stability	Primary tradeoffs
HDI (1,6-hexamethylene diisocyanate, CAS 822-06-0)	Linear C6 aliphatic	Slower; needs DBTDL catalyst	Excellent (non-yellowing)	Highly volatile as monomer (~100,000× MDI vapor pressure); Sage Journals used industrially as biuret/isocyanurate trimers Wikipedia
H12MDI / HMDI (dicyclohexylmethane-4,4'-diisocyanate)	Hydrogenated MDI	Slower than MDI	Excellent	cis/trans isomer mixture; less crystalline hard segments than MDI
IPDI (isophorone diisocyanate)	Asymmetric cycloaliphatic	Two NCO groups with different reactivity	Excellent	Vapor pressure ~100× lower than HDI; Sage Journals gives amorphous hard segments

Aliphatic isocyanates "have no conjugated double bonds and therefore do not interact with light" — the preferred choice for UV-stable PU (ISO-Elektra). Aromatic isocyanates are "significantly faster reacting than aliphatic isocyanates" and "MDI will not need additional catalysts, while strong catalysts such as dibutyltin dilaurate are needed for aliphatic isocyanates" (PCI Magazine, "Isocyanates, Part I"). In a matched five-isocyanate comparison, MDI-based PU exhibits hydrogen-bonding index (HBI) 4.10 and tensile strength 23.4 MPa, the highest of the five; HDI/H12MDI/IPDI give lower TS but improved phase-separation softness and elastomeric character (Liu et al. 2024, PMC11548432, *Polymers*). [ISO-ELEKTRA + 2](#)

The **toxicology case for aliphatic-only is decisive for infant apparel**. Aromatic diisocyanates undergo thermal and physiological hydrolysis releasing 2,4-TDA, 2,6-TDA, and 4,4'-MDA — IARC- and NTP-classified potential human carcinogens: "PUs prepared with TDI underwent degradation under physiological conditions to yield 2,4-toluene diamine (TDA)... MDA has been shown to be mutagenic in *Salmonella typhimorium* test strains, and to cause cancer in both rats and mice, with primary tumor sites being in the liver, kidney, and thyroid" (ScienceDirect 2,4-Diaminotoluene topic page). The EUROPUR CertiPUR program limits aromatic diamines in PU foams to <5 mg/kg (Karlsson et al. 2023). By using HDI/H12MDI/IPDI, Engine A's degradation products are hexamethylene diamine (HDA) and isophorone diamine (IPDA) — strongly basic aliphatic amines that are protonated at physiological pH, water-soluble, and explicitly outside the

1c. Soft-segment chemistry: polyether vs polyester vs polycarbonate

Soft segment	Tg (homopolymer)	Hydrolytic / enzymatic susceptibility	Elastomeric performance
PTMEG (poly-THF, polyether)	$\approx -75\text{ }^{\circ}\text{C}$ (2000 g/mol; Gantrade)	Highly resistant; strain-induced crystallization	Industry standard for spandex; high resiliency
PPG (polyether)	$\approx -65\text{ }^{\circ}\text{C}$	Resistant	Lower mechanical properties; no cold-hardening
PEG (polyether)	$\approx -60\text{ }^{\circ}\text{C}$	Hydrophilic but enzymatically resistant	Less common for elastomers
PCL (polyester)	$\approx -60\text{ }^{\circ}\text{C}$	Hydrolyzable by lipases, cutinases, esterases	Mechanically strong; lipase degradation up to $\sim 97\%$ of PCL electrospun mesh in 90 d (PMC9415937)
Polyadipate esters	varies	Hydrolyzable but poor hydrolytic stability	Cost-sensitive
Polycarbonate diol (Placel CD220)	slightly higher than PCL	Hydrolyzable; $\sim 10\times$ more stable than polyadipates	Equivalent compression set; lower abrasion than PCL

For Engine A, **PCL-diol (2000 g/mol) or polycarbonate-diol soft segments are the correct choice** because they place the cleavable ester (PCL) or carbonate bond directly in the soft segment / amorphous phase, with Tg deep enough ($\approx -60\text{ }^{\circ}\text{C}$) below body temperature for chain mobility. PCL crystallinity ($\sim 40\text{--}60\%$ in bulk; ChemicalBook) is reduced in segmented elastomers due to confinement; the amorphous fraction is what is enzymatically accessible. Park et al. 2023 (Polymers 15:3209) demonstrated PCL-TPE achieving 39.7 MPa UTS and 47.6% biodegradability vs 23.4 MPa and 24.3% for PTMEG analogues. [ChemicalBook](#) [nih](#)

1d. Chain extenders

Standard chain extenders are BDO (1,4-butanediol), EG (ethylene glycol), and EDA (ethylenediamine; the spandex standard, gives urea links). For Engine A, the design space can be expanded with chain extenders carrying built-in cleavable points:

- **Hydroxyethyl disulfide (HEDS)** → dynamic disulfide-bond reshuffling (waterborne PU with PET-derived chain extender, achieving 82.6% strain and 93.7% tensile-strength healing rates; ScienceDirect S2468519423005086).
- **Acetal-bearing diols** → acid-hydrolyzable; closed-loop depolymerization recovers acetaldehyde and MDA from 30 wt% HPC acetal-TPU (Macromolecules 2023, DOI 10.1021/acs.macromol.3c01645). (ACS Publications)
- **Phenol-carbamate dynamic linkers** → thermo-reversible self-healing networks (ScienceDirect S0032386119307943).

These dynamic-bond strategies extend the chemical-recyclability toolbox but are NOT the same as enzymatic cleavage. Engine A should treat them as fallback chemistries should the carbamate-hydrolase enzyme (Engine B) prove too slow on the urethane bond.

1e. Enumeration size and RDKit soundness

- Diisocyanates: HDI, H12MDI, IPDI → 3
- Soft segments: PCL-diol (1000, 1500, 2000, 3000), polycarbonate diol (1000, 2000), PCL-co-PLA (2 ratios) → 8
- Chain extenders: BDO, EG, EDA, HEDS, acetal-BDO → 5
- Hard-segment wt%: 25/27.5/30/32.5/35 → 5
- Base candidates: $3 \times 8 \times 5 \times 5 = 600$

Adding stoichiometric NCO/OH ratios (0.95, 1.00, 1.05) yields ~1,800–4,500 candidates — readily enumerable in RDKit (PSMILES + Stochastic Polymer Representation) in minutes. This is "tractable virtual screening," NOT "billion-scale generative design."

1f. Is a drop-in cleavable elastane chemically plausible?

The peer-reviewed precedent is mixed and honest:

- **Aliphatic-PU TPUs reaching elastane-like elongation:** YES, demonstrated in films/melts. Špírková et al. 2009 (*Polymer*) show all-aliphatic HDI poly(carbonate-urethane)s with EB 600–800%; aliphatic HDI/IPDI poly(ester-carbonate-urea-urethane)s reach TS up to 60 MPa and EB ~500% (Rokicki/Florjańczyk group, ResearchGate 258670101). The C-IP-SS IPDI TPU undergoes reversible strain-induced crystallization at 400% elongation, recovering original strain and transparency on unloading (Eom et al. 2021, *Nat. Commun.* 12:621, DOI 10.1038/s41467-021-20931-z). (ScienceDirect + 2)

- **Aliphatic spun fibers (true fiber form):** NOT demonstrated in peer-reviewed primary literature with simultaneously reported EB + TS + elastic recovery. Patents (Solutia US 7,015,299 for melt-spun aliphatic TPU fibers with EB >400% and tenacity ≥ 0.6 g/denier; Asahi CN1396320A; Lyondell/Bayer US 5,723,563 with EB >900% but using aromatic TDI) describe the territory.
- **Polyester-PU enzymatic degradation:** YES, demonstrated quantitatively (CpCut1, UMG-SP2 — see Part 4).
- **The COMBINATION (aliphatic + polyester soft + cleavable + spun + elastane-grade recovery):** Not yet demonstrated. Engine A's contribution is precisely this co-design.

PART 2 — Structure-Property Relationships

2a. Hard-segment content vs mechanical properties

The literature is unanimous: increasing HS content monotonically (i) raises tensile strength, Young's modulus, hardness, tear strength; (ii) reduces EB and hydrolytic resistance.

- Pugar et al. 2022 (*ACS Appl. Mater. Interfaces* 14:16568, DOI 10.1021/acsami.1c24715): Young's modulus of PU/PUU elastomers correlates strongly with HS fraction; 63-elastomer dataset. [ACS Publications](#)
- Ding et al. 2023 (*Chin. J. Polym. Sci.* 41:422, DOI 10.1007/s10118-022-2838-6): R^2 0.73–0.78 on raw dataset; 0.85–0.91 on benchmark dataset (50.4% of samples retained) for Young's modulus, TS, EB. [Springer](#) [Cjps](#)
- The closest peer-reviewed quantitative example for HS-content-driven TS change is Bian et al. 2024 (*Int. J. Polym. Sci.* DOI 10.1155/2024/9248135): "At a TDI/MDI ratio of 7/3, the tensile strength reaches 52.9 MPa, which is increased by 15.8% than the PUU (10/0)." [Wiley Online Library](#)
- Gantrade technical review of Pechhold-Pruckmayr work: "the elongation at break and hydrolysis-resistance increases with an increase in the PTMEG content." [Gantrade](#)
- PMC7664240 (MDPI Materials 2020): "With identical hard segment content, as r increased, the polyester-based TPUs showed similar physical and mechanical properties as the polyether-based TPUs... tensile strength increased significantly, while the elongation at break decreased." [PubMed Central](#)

Validation of the 25–35 wt% target: This window is squarely in the elastomeric regime where EB > 400% is regularly achieved. Below ~20 wt% HS, hard domains are insufficient for physical crosslinking and recovery; above ~40 wt% the material trends toward leathery/rigid. Engine A's 25–35 wt% target is the correct window for >400% EB. 

2b. The cleavability vs durability tension

The fundamental tradeoff is real and quantifiable: every cleavable bond in the backbone is a potential kinetic weak point under cyclic tensile load. Three mitigations exist:

1. **Place the cleavable bond in the soft segment, NOT the hard segment.** Hard-segment integrity dominates fatigue/recovery; soft-segment scission affects modulus and creep but spares the hard physical crosslinks. This is what Engine A specifies. ✓
2. **Use ester bonds rather than disulfide/acetal in load-bearing positions.** Ester bonds carry mechanical load comparably to ether bonds at room temperature; their lability is largely chemical/enzymatic, not mechanical.
3. **Ensure enzymatic accessibility is in the amorphous phase ONLY** so the hard-segment crystalline lock-in survives normal-wear conditions.

Quantitative sweet spot: A PCL-diol (2000 g/mol) soft segment with 30 wt% H12MDI/BDO hard segment is consistent with EB ~500–700% and TS 15–25 MPa in published PCL-TPE (Park et al. 2023, *Polymers* 15:3209) and is enzymatically degradable at soft-segment ester bonds. This is the plausible target zone.

2c. Crystallinity gates enzyme access

The PET/PETase literature is the gold-standard analog. PETase activity drops ~10-fold when PET crystallinity rises from 2–3% (amorphous film) to ~45% (bottle-grade) (Wei et al. 2022, *Nature Commun.* 13:7180, DOI 10.1038/s41467-022-34908-z; Wang et al. 2023, *ChemBioChem* DOI 10.1002/cbic.202200503). The bioRxiv 2025.12.05 paper makes the molecular case: "In semicrystalline thermoplastics like PET, enzymatic degradation predominantly occurs in the mobile amorphous regions, as the tightly packed chains in crystalline domains are generally inaccessible to the enzyme." [biorxiv](#) [bioRxiv](#)

Spandex itself is highly fatigue-resistant in part because (i) PTMEG soft segments undergo strain-induced crystallization and (ii) MDI/EDA hard domains are semicrystalline. Engine A's design correctly places cleavability in the *amorphous* soft phase, while preserving hard-segment crystallinity for elastic recovery.

2d. Glass transition requirements

For the soft segment to be elastomeric at body temperature, $T_g(\text{soft}) \ll 37\text{ }^{\circ}\text{C}$. Standard targets:

- PTMEG-2000: $T_g \approx -75\text{ }^{\circ}\text{C}$ (Gantrade) ✓
- PCL-diol 2000: $T_g \approx -60\text{ }^{\circ}\text{C}$ (ChemicalBook) ✓
- Polycarbonate diol 2000: T_g slightly higher than PCL, still well below $0\text{ }^{\circ}\text{C}$ (Gantrade) ✓

ML predicts polymer Tg with the best accuracy of any property; PolyID MAE 19.8/26.4 °C (Aubuchon 2023). For soft-segment selection (predicted Tg < -30 °C target), this accuracy is adequate to discriminate elastomeric from non-elastomeric candidates.

2e. Elastic recovery and hysteresis

Reversible >400% extension and snap-back come from (i) entropy elasticity of the amorphous soft segment, (ii) physical crosslinks formed by hard segments, (iii) low hysteresis enabled by low soft-segment Tg and absence of strain-induced soft-segment crystallization that doesn't recover. Introducing cleavable bonds in the soft segment does not, *a priori*, compromise reversible elasticity — unless those bonds have enough conformational restriction to raise Tg. PCL and polycarbonate diols already serve as PU soft segments with high cyclic recovery commercially. **However, NO ML model predicts cyclic recovery directly** — only wet-lab cyclic tensile testing does. Engine A must concede this.

PART 3 — ML Property Predictors: Accuracy, Applicability Domain, and Limits

3a. PolyID (NREL, Aubuchon et al. 2023, *Macromolecules*, DOI 10.1021/acs.macromol.3c00994)

- Architecture: end-to-end multioutput message-passing graph neural network (MPNN); polymers represented from monomer SMILES via the m2p package. (NREL Research Hub)
- Training data: 1,791 curated polymers with up to 8 properties (Tg, Tm, density, modulus, permeabilities for O2, N2, CO2, H2O). (nih)
- Reported accuracy: Tg MAE 19.8 °C on a 20% held-out test set; 26.4 °C on 22 in-house synthesized polymers. (NREL Research Hub)
- Built-in domain-of-validity (DoV) method flags out-of-domain predictions. (NREL Research Hub)
- **Applicability to segmented PU-urea:** PolyID was validated on polyesters and polyamides, NOT on block segmented PU-ureas. For Engine A, PolyID is appropriate for (a) homopolymer Tg of each segment in isolation, (b) ballpark Tg of repeat-unit approximations. The DoV will likely flag segmented PU-urea candidates as out-of-domain — this is a feature, and must be reported transparently. (ACS Publications)

3b. polyBERT (Kuenneth & Ramprasad 2023, *Nat. Commun.* 14:4099, DOI 10.1038/s41467-023-39868-6)

- Architecture: DeBERTa-based encoder-only Transformer trained on 100 million hypothetical PSMILES strings. (NCBI + 3)
- The hypothetical dataset is built by BRICS-decomposing exactly **13,766 known (previously-synthesized) polymers into 4,424 unique fragments**, then enumeratively

recombining (Fig. 1c of the paper). (NCBI)

- Outputs ~29–36 properties via multitask ML on polyBERT fingerprints.
- Speed: ~100× faster than handcrafted Polymer Genome fingerprints. (Georgia Tech News)
- **Honest limit:** polyBERT is trained on hypothetical homopolymers/random-copolymers. The Nature Communications paper itself acknowledges that "polymer types such as homopolymers, copolymer, and polymer blends" need further extension. polyBERT does not natively encode block sequence, hard/soft topology, or microphase morphology. For Engine A, polyBERT can ballpark the bulk properties of an effective average repeat unit but cannot represent the alternating block architecture that is the entire physical basis of spandex behavior. (NCBI)

3c. XGBoost and PU-specific ML models

- **Pugar 2022** (DOI 10.1021/acsami.1c24715): 63-elastomer dataset; physicochemical features + ML bridging molecular → microstructure → bulk scales for Young's modulus. (ACS Publications) (ResearchGate)
- **Ding 2023** (DOI 10.1007/s10118-022-2838-6): R^2 0.73–0.78 on raw dataset; 0.85–0.91 after removing 49.6% of samples through "recursive data elimination" — i.e., $R^2 > 0.85$ only inside a constrained sub-region. (Cjps)
- **Menon, Thompson-Colón & Washburn 2019** (Frontiers in Materials, DOI 10.3389/fmats.2019.00087): hierarchical ML on stress-at-break, strain-at-break, $\tan \delta$ from 18 polymers (training 14, test 4). (Frontiers)
- **2025 PU-Tg XGBoost** (ACS Appl. Polym. Mater., DOI 10.1021/acsapm.5c04524): $R^2 > 0.91$, RMSE ≈ 9.6 K. (ACS Publications)
- **Meng, Lin & Zhang 2024** (Macromolecules 57:10912): MD + ML for multicomponent PU elastomers.

For Engine A: Tg can be predicted with high confidence; modulus and TS with moderate confidence inside the training-data sub-domain; EB with moderate-to-low confidence; **cyclic fatigue, recovery, hysteresis are NOT predictable** by any of these models.

3d. The honest limits — what ML cannot do for PU elastomers

Property	ML predictability today	Why
Tg, Tm	High (MAE ~20 °C)	Many homopolymer data; structural
Density	High	Additive group contributions work
Young's modulus	Moderate within training domain	Pugar 2022; small datasets
Tensile strength	Moderate	Limited datasets
Elongation at break	Moderate-to-low	Microstructure-dependent
Cyclic fatigue	No published model achieves this for PUs	No large dataset; physics is morphology-dependent
Elastic recovery / hysteresis	No published model	Same
Spinnability / processability	No model	Process-dependent
Microphase morphology	Possible via DPD/coarse-grained MD, NOT current ML	Requires real-space simulation

Engine A's claim must be honestly bounded: "predicted Tg + predicted elastomeric class + designed cleavage site placement." Anything stronger requires wet-lab validation.

3e. Why constraint-guided virtual screening > generative design here

Generative polymer design (polyBART, polymer GAN/VAEs) struggles with segmented block copolymers because (i) they lack training data on block architectures; (ii) they cannot represent block sequence and microphase morphology directly; (iii) chemical validity of generated PSMILES is not guaranteed. Constraint-guided enumeration of REAL monomer combinations — RDKit-built, polyBERT/PolyID-scored, with hard chemical constraints (no aromatic, fixed NCO/OH stoichiometry, defined HS wt% bins) — is more defensible because every candidate is (a) chemically real, (b) synthesizable by step-growth polymerization with industrial monomers, (c) interpretable. The Ramprasad group itself describes this as "search synthetically accessible chemical spaces for application-specific polymers at unprecedented scales" (Georgia Tech press release, July 2023). [Georgia Tech News](#)

4a. Which enzymes cleave which bonds

- **Ester bonds (polyester soft segments):** lipases (e.g., *Candida antarctica* lipase B), cutinases (LCC, HiC, CpCut1, ICCG), esterases. Workhorses of polyester-PU degradation. Branson 2023: "the hydrolysis of the ester bonds in polyester-polyurethanes by cutinases is known." (PubMed) (Wiley Online Library)
- **Urethane (carbamate) bonds:** True urethanases UMG-SP1, UMG-SP2, UMG-SP3 (Branson 2023, *Angew. Chem.* 62:e202216220, DOI 10.1002/anie.202216220) hydrolyze the C-O-C(=O)-N moiety using a Ser-SerHis-Lys triad of the amidase signature family (Paiva et al. 2024, *Chem. Sci.* DOI 10.1039/D4SC06688J; rate-limiting acylation step $\Delta G^\ddagger = 20.8$ kcal/mol). (Wiley Online Library + 3)
- **Urea bonds (PU-urea hard segments of spandex):** Ureases. Phua et al. 1987 showed urease degradation of polyether-PU. (ScienceDirect)

For Engine A, the most de-risked enzymatic target is **the polyester ester bond in the soft segment**, attacked by a designed cutinase-class hydrolase. Carbamate hydrolysis in the soft segment by a urethanase is a secondary path piggy-backing on the Branson/Bayer urethanase pipeline.

4b. Precedent for enzymatically degradable PU

- **Pestalotiopsis microspora** (Russell et al. 2011, *Appl. Environ. Microbiol.* 77:6076, DOI 10.1128/AEM.00521-11): endophytic fungi degrading polyester-PU (Impranil DLN) aerobically AND anaerobically; ~21 kDa serine hydrolase implicated. (PubMed +2) (PubMed Central +2)
- **CpCut1** (Liu et al. 2024, *AEM* 90:e01477-23, DOI 10.1128/aem.01477-23): **40.5% TPU film and 20.6% PU foam weight loss in 12 h at 55 °C**; full depolymerization of Impranil DLN. (ResearchGate) (ASM Journals)
- **UMG-SP1/SP2/SP3** (Branson et al. 2023, DOI 10.1002/anie.202216220): hydrolyze a broad range of low-MW dicarbamates from glycolyzed polyether-PU; **UMG-SP2 hydrolyzed >90% of TDA-DEG (20 mM) in 24 h.** (Wiley Online Library + 2)
- **Bayer et al. 2024** (*Angew. Chem.* 63:e202404492, DOI 10.1002/anie.202404492): crystal structure + engineering of UMG-SP-1 yields ~3-fold and ~8-fold activity gains on N-aryl urethane and amide bonds; direct release of MDA from thermoplastic polyester-PU after short incubation times.
- **Li et al. 2025** (*Adv. Sci.* 12:2416019, DOI 10.1002/adv.202416019): **engineered UMG-SP2 A141G shows >30.7-fold improved activity on polyester-PUR** (and Q399A shows 7.4× on an MDA-PUR derivative) vs WT. (nih)

- **Schmidt et al. 2017** (*Polymers* 9:65, PMC6431909) baseline: LCC caused weight losses of "up to 4.9% and 4.1% of Elastollan B85A-10 and C85A-10" thermoplastic polyester-PU cubes in 200 h at 70 °C. [nih](#)

Quantitative gap to PET hydrolases. Comparing the same enzyme class (cutinases) on different substrates: LCC on amorphous PET reaches ~67% depolymerization in ~27 h at 72 °C (~2.5 %/h), vs LCC on polyester-TPU at ~4.9% in 200 h at 70 °C (~0.025 %/h) — a roughly **100-fold per-hour gap** between PET and polyester-TPU enzymatic depolymerization rates. CpCut1's 40.5% in 12 h on TPU film (~3.4 %/h, 2024) closes much of that gap. [PubMed](#)

What makes a PU enzymatically degradable (synthesizing the literature):

1. Polyester (or polycarbonate) — not polyether — soft segment.
2. Amorphous, accessible regions; low overall crystallinity.
3. Hydrophilicity sufficient for enzyme adsorption.
4. Lower molecular weight or lower hard-segment crosslink density.
5. Aliphatic backbone (urethanases work on both aliphatic and aromatic dicarbamates but aliphatic ones lack carcinogenicity baggage).

4c. The central architecture is scientifically correct

Placing a cleavable ester (PCL) or carbamate (urethane) bond in the soft segment, in an accessible amorphous phase, is the design pattern the entire 2011–2025 PU-biodegradation literature converges on. Engine A is on solid ground here. 

4d. Real elastane is highly resistant

Commercial spandex (PTMEG + MDI + EDA) is highly resistant to enzymatic degradation because:

- Polyether soft segments are not enzymatically hydrolyzable in practical time (Branson 2023: "the urethane bonds in polyether-polyurethanes have remained inaccessible to biocatalytic hydrolysis"; Suresh 2024 review: "polyether PU is less susceptible to enzymatic attack"). [PubMed +2](#) [Springer](#)
- MDI/EDA hard segments are semicrystalline and aromatic; aromatic urethanases are slower and bring carcinogenicity baggage.
- Real spandex is also dye- and finish-loaded.

Engine A specifically reverses each of these structural choices: polyester or polycarbonate soft (cleavable) + aliphatic isocyanate (toxicology-clean, also slightly less crystalline hard) + amorphous-accessible bond. The expected enzymatic degradation rate for a redesigned

cleavable fiber should be **orders of magnitude higher** than for commercial spandex, comparable to polyester-TPU rates (CpCut1: 40.5% in 12 h).

4e. The microplastic / end-of-life reframe

This is Engine A's strongest narrative. According to Boucher & Friot (IUCN, 2017, *Primary Microplastics in the Oceans*, Gland, Switzerland) and reaffirmed by the European Environment Agency (2021), **synthetic textiles account for ~35% of primary microplastics released to oceans globally** — making apparel one of the single largest microplastic sources via laundry-wash shedding. End-of-life options for current spandex are: [European Environment Ag...](#)

- Mechanical fragmentation (laundry) → microplastic pollution.
- Incineration → CO₂ + aromatic-amine combustion products.
- Landfill → indefinite persistence.

A redesigned, enzymatically depolymerizable elastane closes the loop: at end of life, garments are treated with the matched carbamate-hydrolase enzyme (Engine B), depolymerizing the fiber to small molecules (HDA + caprolactone + BDO, or analogues) that can be repolymerized into virgin-quality fiber — exactly the V. Tournier (Carbios 2020, *Nature* 580:216) PET model extended to elastane.

PART 5 — Falsification & Red-Team

5a. The strongest objections and rebuttals

Objection 1: "You can't have both >400% stretch AND easy cleavability."

- *Rebuttal:* Demonstrated false in the additive sense. Polyester-TPUs with EB >500% and 47.6% biodegradation exist (Park et al. 2023, *Polymers* 15:3209). Aliphatic HDI poly(carbonate-urethane)s reach EB 600–800% (Špírková 2009). [ScienceDirect](#)
- *Honest concession:* "Easy" is doing work. Polyester-PU TPU enzymatic degradation is still ~10–100× slower per hour than for amorphous PET — Schmidt et al. 2017 (*Polymers* 9:65) found LCC at ~0.025 %/h on polyester-TPU (4.9% in 200 h at 70 °C) versus engineered LCC variants at ~2.5 %/h on amorphous PET (67% in 27 h at 72 °C, Sulaiman et al. 2023). The trade-off is real; correct framing is "engineered cleavability under intentional end-of-life treatment, NOT accidental in-use degradation."

Objection 2: "Aliphatic isocyanates won't give the mechanical performance of MDI spandex."

- *Rebuttal:* Aliphatic PU TPUs reach EB 489–977% and TS 11.07–30.69 MPa (Kim et al. 2022, *J. Polym. Res.* 29:467, DOI 10.1007/s10965-022-03375-x), spanning the spandex range.

Aliphatic patents (US 7,015,299; CN1396320A) describe melt-spun aliphatic PU fibers with EB >400% and tenacity ≥ 0.6 g/denier. (Springer)

- *Honest concession:* Aromatic MDI gives the highest HBI (4.10) and TS (23.4 MPa) in matched tests (PMC11548432, *Polymers* 2024). H12MDI/IPDI give softer hard segments due to ring flexibility and isomer mixing; peak TS may be 20–40% lower than MDI in matched composition. For children's apparel, however, TS is not the binding constraint — comfort, elongation, recovery, and skin safety are. Performance should be characterized as "elastane-class," NOT "MDI-equivalent."

Objection 3: "Your ML models can't predict segmented copolymer properties — the screening is invalid."

- *Rebuttal:* Engine A uses ML for what it CAN do (T_g, density, ballpark modulus, in-domain elastomeric classification) and explicitly DOES NOT use it for what it can't (fatigue, recovery, morphology). PolyID's DoV will flag out-of-domain candidates honestly. The screening's purpose is to PRIORITIZE, not VALIDATE.
- *Honest concession:* polyBERT/PolyID are trained predominantly on homopolymers and random copolymers (polyBERT specifically on BRICS-fragments of 13,766 known homopolymers/simple copolymers); their accuracy on segmented PU-urea is unvalidated. ML scores must be reported with uncertainty bounds and DoV flags, and the top candidates must be flagged for wet-lab validation, not selected as a final product.

Objection 4: "Spinnability / processability isn't captured."

- *Rebuttal:* Correct — and Engine A does not claim it is. Spinnability depends on Mw ($\geq 100,000$ g/mol typical for spandex), solution viscosity, solvent (DMAc/DMF, 25–40 wt% solids for dry-spinning per Bayer patent US 5,057,260), and spinneret geometry. These are wet-lab variables. (Google Patents + 2)
- *Honest concession:* This is a legitimate gap. Engine A's outputs are CANDIDATE CHEMISTRIES, not validated FIBERS.

Objection 5: "Real elastane works because of polyether soft segments — a polyester-soft fiber won't perform like spandex."

- *Rebuttal:* Polyester-PU TPEs match or exceed polyether-PU TPEs in mechanical strength (PCL-TPE UTS 39.7 MPa vs PTMEG-TPE 23.4 MPa; Park et al. 2023). Historical preference for polyether in spandex is driven by hydrolytic stability for everyday washing, NOT for elastomeric performance. PCL and polycarbonate diols give comparable or better tensile and elastic recovery (Gantrade Placel CD220 technical data).
- *Honest concession:* Hydrolytic stability under normal laundry conditions is a real concern. Polycarbonate-diol soft segments (more hydrolytically stable than polyadipates) are the

safer choice; the cleavable bond's design must require enzymatic activation, NOT spontaneous hydrolysis at 40 °C.

Objection 6: "Microphase separation morphology, not just composition, determines properties — you can't screen for it."

- *Rebuttal:* True. Engine A's response is to use composition + topology as proxies (HS wt%, soft-segment MW, segment Tg gap = Flory-Huggins surrogate) and explicitly note that morphology prediction requires DPD or coarse-grained MD, NOT current polymer ML.
- *Honest concession:* This is the limit of the ML approach. The candidate shortlist needs morphology characterization (SAXS/TEM) before any fiber claim.

5b. Genuine open scientific risks

- Will the enzyme (Engine B) hydrolyze the designed cleavable bond at a USEFUL rate in a finished garment matrix (with dyes, finishes, fabric structure)?
- Will the redesigned fiber's elastic recovery survive 1,000+ wash cycles?
- Will polyester soft-segment chemistry survive normal-wear humidity and 40 °C wash water for the garment's intended lifetime?
- Will spinnability be achievable from the prioritized candidates?
- Will depolymerization products be recoverable in industrially useful purity?

5c. How to honestly bound the claims

Engine A CAN claim:

- A prioritized, RDKit-enumerated shortlist of 5–10 candidate fiber chemistries.
- Each candidate has a predicted Tg (PolyID MAE ~25 °C) consistent with elastomeric behavior.
- Each candidate has a designed cleavable bond placed in the amorphous soft segment by chemical construction.
- Each candidate uses only aliphatic isocyanates (HDI, H12MDI, IPDI), eliminating MDA/TDA carcinogenicity risk.
- Each candidate is in the 25–35 wt% HS regime documented to give >400% EB in PU elastomers.

Engine A CANNOT claim:

- Validated mechanical performance (TS, EB, recovery, fatigue) without wet-lab testing.
- Validated enzymatic degradation rate without enzyme assays on the actual polymer.
- Spinnability or processability.

- Microphase morphology.
 - Drop-in equivalence to commercial spandex (Lycra).
 - A regulatory or safety dossier for infant skin contact (separate body of work).
-

Recommendations

Staged next steps

1. **Now (in-silico):** Run RDKit enumeration of the ~600-1,800 candidate chemistries. Score each with PolyID for Tg, polyBERT for ballpark bulk properties, and an XGBoost-PU model (Pugar 2022/Ding 2023 if reproducible, else retrain on PoLyInfo PU subset) for modulus and EB. Filter on: predicted Tg(soft) < -30 °C; HS wt% ∈ [25, 35]; aliphatic isocyanate; PCL- or PC-diol soft segment; BDO/EDA chain extender (HEDS/acetal as fallback). Output top 10 with explicit uncertainty bands and DoV flags.
2. **Next 3 months (wet-lab Tier 1):** Synthesize the top 3 candidates (~100 g batches) by two-step prepolymer method. DSC for Tg/Tm; tensile testing for EB/TS/modulus; SAXS for microphase morphology; enzymatic susceptibility assay with CpCut1 or candidate Engine-B enzyme.
3. **Next 12 months (wet-lab Tier 2):** Best candidate → dry-spinning trial; characterize fiber EB, recovery, cyclic fatigue (100+ cycles to 300% strain); enzymatic depolymerization to monomers in pilot reactor.
4. **Thresholds to change strategy:** If the top candidate after Tier 1 fails any of (Tg(soft) < 0 °C, EB > 350%, ≥10% weight loss with selected enzyme in 24 h at 40 °C, distinct two-Tg DSC indicating phase separation), pivot — most likely to polycarbonate-diol soft (more hydrolytically stable, still cleavable) or higher-Mw PCL.

Prioritized candidate shortlist (top 5)

Rank	Diisocyanate	Soft segment	Chain extender	HS wt%	Rationale
1	H12MDI	PCL-diol 2000 g/mol	BDO	30%	Closest known analog: Eceiza et al. 2021 (Polymers 13:2060, DOI 10.3390/polym13132060) H12MDI/HDO/PCL PU; (MDPI) PCL is enzymatically labile; H12MDI gives MDI-like symmetry without aromaticity.

Rank	Diisocyanate	Soft segment	Chain extender	HS wt%	Rationale
2	HDI	Polycarbonate diol 2000 (Placel CD220-like)	BDO	27.5%	Špírková et al. 2009 all-aliphatic poly(carbonate-urethane) with EB 600–800%; ScienceDirect PCD gives ~10× hydrolytic stability vs polyadipates; carbonate bond is enzymatically cleavable but slower than ester (controlled-rate design).
3	IPDI	PCL-diol 2000	EDA	30%	EDA restores urea-bond H-bonding for spandex-class recovery (cf. Eom 2021 C-IP-SS); IPDI gives amorphous hard segments tunable by isomer ratio.
4	H12MDI	PCL-co-PLA diol 2000 (80:20)	BDO	30%	Faster enzymatic degradation than pure PCL; intermediate Tg; larger amorphous fraction.
5	HDI	PCL-diol 2000	HEDS (hydroxyethyl disulfide)	28%	Dual-cleavage fallback (enzymatic ester + chemo-cleavable disulfide); reducing-environment depolymerization as backup if enzyme is too slow.

Key citations (DOIs) for the submission

1. Castagna, Pangon, Choi, Dillon, Runt 2012 — *Macromolecules* 45:8438–8444. DOI 10.1021/ma3016568.
2. Kuenneth & Ramprasad 2023 — *Nat. Commun.* 14:4099. DOI 10.1038/s41467-023-39868-6. (polyBERT)
3. Aubuchon et al. 2023 — *Macromolecules*. DOI 10.1021/acs.macromol.3c00994. (PolyID)
4. Pugar, Gang, Huang, Haider, Washburn 2022 — *ACS Appl. Mater. Interfaces* 14:16568–16581. DOI 10.1021/acsami.1c24715.
5. Ding, Liu, Liu, Li, Li, Sun 2023 — *Chin. J. Polym. Sci.* 41:422–431. DOI 10.1007/s10118-022-2838-6.
6. Liu, Xin, Zhang et al. 2024 — *Appl. Environ. Microbiol.* 90:e01477-23. DOI 10.1128/aem.01477-23. (CpCut1: 40.5%/20.6% in 12 h)
7. Branson, Sötl, Buchmann, Wei, Schaffert, Badenhorst, Reisky, Jäger, Bornscheuer 2023 — *Angew. Chem. Int. Ed.* 62:e202216220. DOI 10.1002/anie.202216220. (UMG-SP1/SP2/SP3)

8. **Bayer, Palm, Berndt et al. 2024** — *Angew. Chem. Int. Ed.* 63:e202404492. DOI 10.1002/anie.202404492. (Engineered UMG-SP-1)
 9. **Li et al. 2025** — *Adv. Sci.* 12:2416019. DOI 10.1002/advs.202416019. (UMG-SP2 A141G 30.7-fold)
 10. **Paiva, Teixeira et al. 2024** — *Chem. Sci.* 16:2437–2452. DOI 10.1039/D4SC06688J. (UMG-SP2 mechanism; $\Delta G^\ddagger$ 20.8 kcal/mol)
 11. **Russell et al. 2011** — *Appl. Environ. Microbiol.* 77:6076–6084. DOI 10.1128/AEM.00521-11. (*Pestalotiopsis microspora*)
 12. **Wei, Cui et al. 2022** — *Nat. Commun.* 13:7180. DOI 10.1038/s41467-022-34908-z. (PET crystallinity-degradation)
 13. **Park et al. 2023** — *Polymers* 15:3209. DOI 10.3390/polym15153209. (PCL-TPE 47.6% biodegradation, UTS 39.7 MPa)
 14. **Spence & Plehiers 2022** — *Toxicol. Ind. Health* 38. DOI 10.1177/07482337221090709. (Diisocyanate toxicology)
 15. **Karlsson, Spence, Plehiers 2023** — *Toxicol. Ind. Health* 39:298–312. DOI 10.1177/07482337231172816. (TDA/MDA migration)
 16. **Eom, Jeon et al. 2021** — *Nat. Commun.* 12:621. DOI 10.1038/s41467-021-20931-z. (IPDI-TPU strain crystallization at 400% EB)
 17. **Špírková et al. 2009** — *Polymer*. DOI 10.1016/j.polymer.2009.03.034. (All-aliphatic HDI poly(carbonate-urethane) EB 600–800%)
 18. **Kim et al. 2022** — *J. Polym. Res.* 29:467. DOI 10.1007/s10965-022-03375-x. (H12MDI bio-TPU EB 489–977%)
 19. **Eceiza et al. 2021** — *Polymers* 13:2060. DOI 10.3390/polym13132060. (H12MDI/HDO/PCL-PU)
 20. **Schmidt et al. 2017** — *Polymers* 9:65. PMC6431909. (LCC on Elastollan polyester-TPU, 4.9% in 200 h)
 21. **Tournier et al. 2020** — *Nature* 580:216–219. (Engineered PET depolymerase, ICCG)
 22. **Liu et al. 2024** — *Polymers* PMC11548432. (Five-isocyanate matched comparison; MDI HBI 4.10, TS 23.4 MPa)
 23. **Bian et al. 2024** — *Int. J. Polym. Sci.* DOI 10.1155/2024/9248135. (TDI/MDI ratio 7/3, TS +15.8% vs 10/0)
 24. **Boucher & Friot 2017** — IUCN, *Primary Microplastics in the Oceans* (Gland, Switzerland). (Synthetic textiles $\approx$ 35% of primary ocean microplastics)
 25. **2025 PU-Tg XGBoost** — *ACS Appl. Polym. Mater.* DOI 10.1021/acsapm.5c04524. ($R^2 > 0.91$, RMSE 9.6 K)
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Caveats

1. **No peer-reviewed primary publication of a true dry-spun HDI-only / H12MDI-only / IPDI-only spandex filament** with simultaneously reported elongation, tensile strength, AND elastic recovery values was located in the searched literature. Commercial spandex remains aromatic-MDI/TDI based; aliphatic-only spandex appears only in patent claims (US 7,015,299; CN1396320A) and industry/bio-based press (Hyosung). Engine A's "aliphatic spandex" is therefore genuinely novel territory, NOT a re-discovery.
 2. **Absolute kcat/Km values** for current PU urethanases (UMG-SP1/SP2/SP3) on polymer (not low-MW dicarbamate) substrates, in $\text{s}^{-1}\cdot\text{M}^{-1}$, were not located in the literature searched. The consensus is qualitative: PU enzymatic degradation is "orders of magnitude" slower than PET enzymatic degradation today (Bayer 2024: "polyurethanes (PUs) and polyamides (PAs) largely resist biodegradation"). A specific quantitative anchor: Schmidt et al. 2017 LCC on Elastollan polyester-TPU gives $\sim 0.025\text{ \%}/\text{h}$ vs engineered LCC on amorphous PET $\sim 2.5\text{ \%}/\text{h}$ — roughly $100\times$ per-hour gap. (Wiley Online Library)
 3. **PolyID, polyBERT, and PU-XGBoost models** are all trained predominantly on homopolymer and random-copolymer data. polyBERT's training set is built by BRICS-fragmenting exactly 13,766 known polymers into 4,424 fragments and recombining; segmented block PU-ureas are outside the validated domain. (arXiv)
 4. **Cyclic fatigue, true elastic recovery, hysteresis, spinnability, and microphase morphology** are not predicted by any current polymer ML model. These are the highest-priority wet-lab measurements.
 5. **PCL soft-segment crystallinity** (40–60% in bulk) can reduce enzymatic accessibility; lower-MW PCL diols (1000–2000 g/mol) in segmented PU give lower crystallinity due to confinement, but this remains a tuning variable.
 6. **Polyester (PCL) hydrolytic stability under laundry conditions (40 °C, detergent)** is a real concern; the cleavable bond's design must require enzymatic activation, NOT spontaneous hydrolysis during ordinary wear.
 7. **All conclusions are in-silico and literature-based.** No wet-lab synthesis or fiber characterization was performed. Engine A's deliverable is a SHORTLIST + RATIONALE, not a validated material.
 8. **The "12% TS / 760%→680% EB at 40→45% MDI"** figure cited in industry sources is not located in peer-reviewed primary literature; the closest peer-reviewed quantitative example is Bian et al. 2024 reporting "15.8% TS increase" at a TDI/MDI 7/3 ratio. The direction of the trend (higher HS → higher TS, lower EB) is robustly peer-reviewed (Pugar 2022; Ding 2023; MDPI Materials 13:4991 2020) but exact percentages depend on formulation.
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Final assessment

Is Engine A chemically plausible and defensible?

YES, as a constraint-guided in-silico shortlisting exercise with a clear handoff to wet-lab validation. Every design constraint Engine A locks in (aliphatic-only diisocyanate; 25–35 wt% HS; polyester or polycarbonate soft segment with cleavable bond in amorphous phase; aliphatic chain extender) is independently supported by peer-reviewed primary literature. The combination is genuinely novel — no dry-spun, all-aliphatic, enzymatically depolymerizable spandex filament has been published — but each piece is established. The honest scientific framing of "predicted Tg + predicted elastomeric class + designed cleavage site, NOT validated fiber" is exactly the framing that survives expert scrutiny.

The strongest evidence FOR Engine A:

- Aliphatic PU TPUs reaching EB 600–800% are documented (Špírková 2009; Kim 2022).
- Polyester-PU enzymatic degradation reaching 40.5% in 12 h is documented (Liu 2024, CpCut1).
- Carbamate-bond enzymatic hydrolysis on real PU substrates with engineered urethanases reaching >30-fold WT activity is documented (Li 2025).
- The toxicology case for aliphatic-only in infant apparel is settled (Karlsson 2023; CertiPUR <5 mg/kg).
- The HS-wt% window 25–35% for elastomeric >400% EB is robustly characterized (Pugar 2022; Ding 2023).
- ML can predict Tg of soft-segment chemistry with MAE 19.8–26.4 °C (PolyID) — adequate for elastomeric-class screening.

The strongest evidence AGAINST a strong claim:

- No spun fiber with simultaneous aliphatic-only + cleavable + elastane-grade recovery exists in published literature.
- ML models do not predict cyclic fatigue, elastic recovery, or spinnability.
- Polyester soft-segment hydrolytic stability under repeated washing is a genuine engineering risk.
- Polyester-PU enzymatic degradation is still ~100× slower per hour than PET enzymatic degradation (Schmidt 2017 vs Sulaiman 2023 baseline).

The single most rigorous expert objection is the cyclic-fatigue/recovery question, and Engine A correctly concedes it. **The second most rigorous objection is spinnability**, and Engine A correctly concedes it. With these honest concessions, the prioritized shortlist is defensible as a

starting point for a real R&D program — and the underlying scientific logic (place enzymatically labile bonds in the amorphous soft segment of an aliphatic segmented PU-urea, with HS content tuned for elastane-class elongation) is one of the most sensible reformulations of spandex anyone could propose for the children's apparel use case.